

Associative Memory and Fast Weights: what a 2x2 matrix teaches about frontier memory architectures

Concept summary — “Memory Under Pressure”, DataForge 2026 Pathway track. Artifact & sources: github.com/jainraunak944/dataforge (branch `claude/memory-under-pressure-p6vp89`).

DEFINITION & DESIGN PRESSURE. A fast-weight associative memory stores relationships in rapidly changing connection strengths and retrieves them when a similar cue returns. A Transformer instead appends every token to a key–value cache that grows with context and is re-scanned each step; the pressure is to replace that growing cache with a fixed-size state updated incrementally. The minimal mechanism, with column vectors $k \in \mathbb{R}^{d_k}$, $v \in \mathbb{R}^{d_v}$, state $M \in \mathbb{R}^{(d_v \times d_k)}$, is the additive write $M_t = \lambda M_{(t-1)} + v_t k_t^\top$ with linear read $M q$ — exactly the memory a linear-attention layer maintains [3, Eq. 17].

Our falsifiable claim: storing two associations with orthogonal unit keys in this memory permits exact recall; increasing key overlap ρ introduces calculable interference — $\text{retrieved}(k_A) = v_A + \rho v_B$, L2 error exactly ρ — while the matrix keeps constant shape and no trained parameters change.

MECHANISM & FAILURE MODE. Writes superimpose rank-1 stamps; there are no slots. Reading returns every stored value weighted by cue–key similarity, so overlapping keys contaminate recall; at $\rho=1$ two associations become cue-indistinguishable (our artifact reports the tie rather than fabricating a winner). Capacity is bounded by key dimension: exceeding it forces overlap and guarantees retrieval error [3, §4.1]. At scale this bites: with unlimited context on WikiText-103, a purely additive fast-weight model degrades beyond perplexity 260 while an interference-correcting variant of equal size reaches 29.4 [3, Table 4].

THE 2024–26 RESPONSES. DeltaNet replaces the blind add with an error-correcting write — one gradient step on recall error — precisely because additive writes lead to key “collisions” [1, §2.2]; Gated DeltaNet adds learned decay for fast erasure [2, §3.1]. Both are competitive 1.3B-parameter LMs with open code.

System	Update per token/demo	State	Key evidence (all author-reported)	Trades away
Additive fast weights	$S \ += \ v \ k^\top$	constant	formal capacity bound [3, §4.1]	recall purity under overlap
DeltaNet [1]	$S \ -= \ \beta(Sk - v) \ k^\top$, insert	constant	1.3B: avg zero-shot 51.6 vs Mamba 50.0 [T.1]	no decay → length generalization
Gated DeltaNet [2]	delta + decay α	constant	S-NIAH-2@4K 92.2 vs DeltaNet 18.6 [T.2]	retention (91.8 vs 98.8 @8K)
BDH-GPU [4]	$\rho \ := \ (\rho + \text{LN}(E_y) x^\top) U$ (Eq. 8)	constant	GPT-2-class LM at 10M–1B [Fig. 7]	maturity; no external replication
BDH-CQ [5]	$S_t = U_\theta(S_{(t-1)}, D_t)$ (Eq. 1)	constant	ARC-AGI-1 29.50% pass@2, $\approx \$0.0007/\text{task}$ [T.1, §5]	accuracy vs top LLMs; internals proprietary

WHERE BDH AND BDH-CQ FIT. BDH reformulates attention as synaptic memory: BDH-GPU carries per-layer fast state ρ , written as the outer product of a learned value $\text{LN}(E_y)$ with the current sparse non-negative activation x (serving as key *and* query), read as ρx , decayed by an operator U generalizing our λ [4, §3.2, Eq. 8]. Trained matrices E , D_x , D_y stay frozen at inference — the state/parameter split our toy isolates. BDH is not a Mamba-style SSM; the paper derives it from local neuron–synapse rules, with linear attention in a high-dimensional neuronal space as its tensor form [4, §3]. BDH-CQ lifts the same split to task acquisition: contextual memory accumulates per demonstration (additive accumulation is its stated linear-attention special case [5, §3.2]), then a latent workspace iterates against the frozen memory — reasoning without chain-of-thought tokens and, explicitly, “no parameters are updated at inference time” [5, §1]. The roles are unequal: BDH supplies the mechanism our toy maps onto; BDH-CQ the system-level evidence that state-based adaptation works.

EVIDENCE, LABELED. All numbers above are developer-reported. BDH-CQ’s 118/400 (≈ 0.85 H200 GPU-s/task) was audited black-box — by co-authors, hence semi-independent; its report discloses proprietary internals and a training-overlap caveat [5, §3.3, §6.5]. Pathway’s AWS work and 1B–600B scaling are company announcements, not evaluations. No system in the table has an independent external reproduction we could verify — the field’s most important missing evidence. Our maturity assessment (0–10; 10 = standardized infrastructure; ours, not official): delta-rule linear attention ≈ 6 ; BDH/BDH-CQ ≈ 3 .

THE SUBMISSION. An interactive essay where the claim is testable in under a minute: one overlap slider drives a live 2x2 memory with ground truth beside retrieval, an independently coded explicit-history reference (tolerance 10^{-9}) prevents self-confirming code, cells decompose to show where interference is born, and a sourced BDH/BDH-CQ equation walkthrough asks, symbol by symbol, what changes during inference. The open question the learner leaves with: how should useful fast state consolidate into slow weights — unsolved in every system above.

[1] Yang et al., Parallelizing Linear Transformers with the Delta Rule over Sequence Length, NeurIPS 2024, arXiv:2406.06484 · [2] Yang, Kautz, Hatamizadeh, Gated Delta Networks, ICLR 2025, arXiv:2412.06464 · [3] Schlag, Irie, Schmidhuber, Linear Transformers Are Secretly Fast Weight Programmers, ICML 2021, arXiv:2102.11174 (background; pre-2022) · [4] Kosowski et al., The Dragon Hatchling, 2025, arXiv:2509.26507 · [5] Engdahl et al., BDH-CQ: In-Context Learning with Recurrent Latent Reasoning, 2026, arXiv:2608.09888. Equation-level ledger: docs/RESEARCH.md in the repo. Continue: [4] §3, [5] §3, Pathway’s “Equations of Reasoning”.